No.	Status	Date
LNDE-RPT-96-014-DS-6-LA-103	Issued	1996-03-08

UT Thickness Data Sheet - 6-LA-103

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for the 6-LA-103 lean amine header located within the ATU-100. The objective is to establish baseline wall thickness data for the line prior to the unit entering service.

Grace Okafor, acting as the UT technician, performed the measurements. The technician is certified to SNT-TC-1A UT Level II through Larkhall NDE Services Inc.

Technical Basis

All thickness measurements were performed under the requirements of QSCP-NDT-001 Rev 0. The inspection methodology adheres to the guidelines specified in ASTM E797.

The data recorded herein supports the findings detailed in LNDE-RPT-96-014, which covers the baseline survey of 118 CMLs for the ATU-100.

Component Specifications

The subject line, 6-LA-103, is constructed from A106 Gr.B with a nominal outside diameter of 168.3 mm and a nominal wall thickness of 7.11 mm.

The design limits for this component are 2413 kPa(g) and 93 °C. The minimum allowable thickness (t-min) is 3.56 mm.

Evaluation Criteria

Acceptance is based on the retirement thickness, which is defined as the t-min of 3.56 mm. Any reading measured below this value shall be raised as an NCR.

The 6-LA-103 line is categorized under Class 2 for amine service, which dictates a maximum thickness inspection interval of 10 years and an external visual interval of 5 years.

Thickness readings, measured 1996-03-08

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	—	—	7.12	—	—	—	2001-03-08	
6-SG-101-02	7.11	3.56	—	—	7.05	—	—	—	2001-03-08	
6-SG-101-03	7.11	3.56	—	—	7.08	—	—	—	2001-03-08	
6-SG-101-04	7.11	3.56	—	—	7.09	—	—	—	2001-03-08	
6-SG-101-05	7.11	3.56	—	—	7.14	—	—	—	2001-03-08	
6-SG-101-06	7.11	3.56	—	—	7.07	—	—	—	2001-03-08	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	—	—	7.14	—	—	—	2001-03-08	
6-SG-102-02	7.11	3.56	—	—	7.11	—	—	—	2001-03-08	
6-SG-102-03	7.11	3.56	—	—	7.04	—	—	—	2001-03-08	
6-SG-102-04	7.11	3.56	—	—	6.99	—	—	—	2001-03-08	
6-SG-102-05	7.11	3.56	—	—	7.13	—	—	—	2001-03-08	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	—	—	8.20	—	—	—	2006-03-08	
8-TG-101-02	8.18	4.06	—	—	8.20	—	—	—	2006-03-08	
8-TG-101-03	8.18	4.06	—	—	8.23	—	—	—	2006-03-08	
8-TG-101-04	8.18	4.06	—	—	8.10	—	—	—	2006-03-08	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-05	8.18	4.06	—	—	8.22	—	—	—	2006-03-08	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	—	—	5.99	—	—	—	2006-03-08	
4-LA-101-02	6.02	3.18	—	—	6.01	—	—	—	2006-03-08	
4-LA-101-03	6.02	3.18	—	—	5.96	—	—	—	2006-03-08	
4-LA-101-04	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-101-05	6.02	3.18	—	—	6.01	—	—	—	2006-03-08	
4-LA-101-06	6.02	3.18	—	—	6.06	—	—	—	2006-03-08	
4-LA-101-07	6.02	3.18	—	—	5.99	—	—	—	2006-03-08	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	—	—	6.02	—	—	—	2006-03-08	
4-LA-102-02	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-102-03	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-102-04	6.02	3.18	—	—	5.99	—	—	—	2006-03-08	
4-LA-102-05	6.02	3.18	—	—	6.00	—	—	—	2006-03-08	
4-LA-102-06	6.02	3.18	—	—	6.00	—	—	—	2006-03-08	
4-LA-102-07	6.02	3.18	—	—	6.00	—	—	—	2006-03-08	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	—	—	7.02	—	—	—	2006-03-08	
6-LA-103-02	7.11	3.56	—	—	7.19	—	—	—	2006-03-08	
6-LA-103-03	7.11	3.56	—	—	7.08	—	—	—	2006-03-08	
6-LA-103-04	7.11	3.56	—	—	7.17	—	—	—	2006-03-08	
6-LA-103-05	7.11	3.56	—	—	7.19	—	—	—	2006-03-08	
6-LA-103-06	7.11	3.56	—	—	7.16	—	—	—	2006-03-08	
6-LA-103-07	7.11	3.56	—	—	7.18	—	—	—	2006-03-08	
6-LA-103-08	7.11	3.56	—	—	7.04	—	—	—	2006-03-08	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	—	—	8.52	—	—	—	2006-03-08	
4-RA-101-02	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-101-03	8.56	3.81	—	—	8.61	—	—	—	2006-03-08	
4-RA-101-04	8.56	3.81	—	—	8.63	—	—	—	2006-03-08	
4-RA-101-05	8.56	3.81	—	—	8.60	—	—	—	2006-03-08	
4-RA-101-07	8.56	3.81	—	—	8.60	—	—	—	2006-03-08	
4-RA-101-08	8.56	3.81	—	—	8.68	—	—	—	2006-03-08	
4-RA-101-09	8.56	3.81	—	—	8.57	—	—	—	2006-03-08	
4-RA-101-10	8.56	3.81	—	—	8.67	—	—	—	2006-03-08	
4-RA-101-11	8.56	3.81	—	—	8.68	—	—	—	2006-03-08	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-102-02	8.56	3.81	—	—	8.57	—	—	—	2006-03-08	
4-RA-102-03	8.56	3.81	—	—	8.56	—	—	—	2006-03-08	
4-RA-102-04	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-102-05	8.56	3.81	—	—	8.55	—	—	—	2006-03-08	
4-RA-102-06	8.56	3.81	—	—	8.51	—	—	—	2006-03-08	
4-RA-102-07	8.56	3.81	—	—	8.58	—	—	—	2006-03-08	
4-RA-102-08	8.56	3.81	—	—	8.49	—	—	—	2006-03-08	
4-RA-102-09	8.56	3.81	—	—	8.55	—	—	—	2006-03-08	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	—	—	11.00	—	—	—	2006-03-08	
6-RA-103-02	10.97	4.57	—	—	11.09	—	—	—	2006-03-08	
6-RA-103-03	10.97	4.57	—	—	10.99	—	—	—	2006-03-08	
6-RA-103-04	10.97	4.57	—	—	10.98	—	—	—	2006-03-08	
6-RA-103-05	10.97	4.57	—	—	10.95	—	—	—	2006-03-08	
6-RA-103-06	10.97	4.57	—	—	11.01	—	—	—	2006-03-08	
6-RA-103-07	10.97	4.57	—	—	10.99	—	—	—	2006-03-08	
6-RA-103-08	10.97	4.57	—	—	10.97	—	—	—	2006-03-08	
6-RA-103-09	10.97	4.57	—	—	11.01	—	—	—	2006-03-08	
6-RA-103-10	10.97	4.57	—	—	11.02	—	—	—	2006-03-08	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	—	—	7.59	—	—	—	2001-03-08	
3-RV-101-02	7.62	3.56	—	—	7.57	—	—	—	2001-03-08	
3-RV-101-03	7.62	3.56	—	—	7.52	—	—	—	2001-03-08	
3-RV-101-04	7.62	3.56	—	—	7.59	—	—	—	2001-03-08	
3-RV-101-05	7.62	3.56	—	—	7.67	—	—	—	2001-03-08	
3-RV-101-06	7.62	3.56	—	—	7.59	—	—	—	2001-03-08	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	—	—	5.53	—	—	—	2001-03-08	
2-RV-102-02	5.54	2.79	—	—	5.59	—	—	—	2001-03-08	
2-RV-102-03	5.54	2.79	—	—	5.48	—	—	—	2001-03-08	
2-RV-102-04	5.54	2.79	—	—	5.58	—	—	—	2001-03-08	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	—	—	5.48	—	—	—	2006-03-08	
3-RF-101-02	5.49	2.41	—	—	5.45	—	—	—	2006-03-08	
3-RF-101-03	5.49	2.41	—	—	5.45	—	—	—	2006-03-08	
3-RF-101-04	5.49	2.41	—	—	5.46	—	—	—	2006-03-08	
3-RF-101-05	5.49	2.41	—	—	5.45	—	—	—	2006-03-08	
3-RF-101-06	5.49	2.41	—	—	5.52	—	—	—	2006-03-08	
3-RF-101-07	5.49	2.41	—	—	5.42	—	—	—	2006-03-08	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	—	—	3.92	—	—	—	2006-03-08	
2-RF-102-02	3.91	2.16	—	—	3.88	—	—	—	2006-03-08	
2-RF-102-03	3.91	2.16	—	—	3.93	—	—	—	2006-03-08	
2-RF-102-04	3.91	2.16	—	—	3.87	—	—	—	2006-03-08	
2-RF-102-07	3.91	2.16	—	—	3.96	—	—	—	2006-03-08	

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-104-02	6.02	3.18	—	—	6.03	—	—	—	2006-03-08	
4-LA-104-03	6.02	3.18	—	—	5.96	—	—	—	2006-03-08	
4-LA-104-04	6.02	3.18	—	—	6.06	—	—	—	2006-03-08	
4-LA-104-05	6.02	3.18	—	—	6.04	—	—	—	2006-03-08	
4-LA-104-06	6.02	3.18	—	—	6.10	—	—	—	2006-03-08	
4-LA-104-07	6.02	3.18	—	—	6.01	—	—	—	2006-03-08	
4-LA-104-08	6.02	3.18	—	—	5.94	—	—	—	2006-03-08	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	—	—	8.51	—	—	—	2006-03-08	
4-RA-104-02	8.56	3.81	—	—	8.55	—	—	—	2006-03-08	
4-RA-104-03	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-104-04	8.56	3.81	—	—	8.59	—	—	—	2006-03-08	
4-RA-104-05	8.56	3.81	—	—	8.60	—	—	—	2006-03-08	
4-RA-104-06	8.56	3.81	—	—	8.59	—	—	—	2006-03-08	
4-RA-104-07	8.56	3.81	—	—	8.59	—	—	—	2006-03-08	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	—	—	3.93	—	—	—	2006-03-08	
2-ST-101-02	3.91	2.29	—	—	3.90	—	—	—	2006-03-08	
2-ST-101-03	3.91	2.29	—	—	3.94	—	—	—	2006-03-08	
2-ST-101-04	3.91	2.29	—	—	3.82	—	—	—	2006-03-08	
2-ST-101-05	3.91	2.29	—	—	3.91	—	—	—	2006-03-08	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	—	—	5.42	—	—	—	2006-03-08	
3-CW-101-02	5.49	2.79	—	—	5.41	—	—	—	2006-03-08	
3-CW-101-03	5.49	2.79	—	—	5.53	—	—	—	2006-03-08	
3-CW-101-04	5.49	2.79	—	—	5.46	—	—	—	2006-03-08	
3-CW-101-05	5.49	2.79	—	—	5.48	—	—	—	2006-03-08	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	—	—	5.50	—	—	—	2006-03-08	
3-CW-102-02	5.49	2.79	—	—	5.51	—	—	—	2006-03-08	
3-CW-102-03	5.49	2.79	—	—	5.53	—	—	—	2006-03-08	
3-CW-102-04	5.49	2.79	—	—	5.51	—	—	—	2006-03-08	

Sign-off	Name	Function	Date	Signature
Prepared	Grace Okafor	UT technician, SNT-TC-1A UT Level II	1996-03-08	_____